

A Comparative Analysis of Machine Learning Models for Predicting Student Performance

Reine Nassour
Department of Computer Science
Lebanese American University
reinenassour341@gmail.com

Abstract

Prediction of student academic performance has gained prominence in the domain of education mining and artificial intelligence in the past few years. Identifying students at risk of having poor academic performance is crucial in ensuring proper intervention measures are taken. Machine learning algorithms have been extensively applied over the years to study and predict student performance by analyzing various academic and behavioral parameters of students.

The study examines the efficiency of different machine learning algorithms for predicting the performance of students. Regression models such as Linear Regression, Decision Trees, Random Forest, KNN, Gradient Boosting, Extra Trees, XGBoost, and SVR were utilized and tested for their efficacy. The models were tested against a dataset of student performances characterized by several features.

Coefficients of determination, R^2 values, were used to assess the predictive power of the machine learning algorithms in the analysis of the student performance data set. The results of the experiments reveal a comparative analysis of the algorithms. The models with highest predictive performance are also highlighted.

Keywords

Machine Learning, Student Performance Prediction, Educational Data Mining, Regression Models, Artificial Intelligence

1. Introduction

Predictive modeling of student success or failure has become an important topic of research in the area of educational data mining and analytics. Educational institutions constantly strive for efficient ways of tracking students' development and detecting students who are likely to have difficulties with their studies. Early predictions of student success would give teachers the possibility to take timely action in order to change the learning strategy and improve academic results. Due to the fast increase in educational data produced by learning environments, machine learning models have become popular methods of analysis for this kind of data.

Machine learning methods are being extensively used in recent years for predicting student success or failure using different features of learners. Machine learning algorithms allow building highly efficient predictive models, which are capable of discovering dependencies between different variables, thus providing researchers with valuable information about future performance of individual learners.

Numerous researchers have considered various machine learning approaches for predicting student performance. Methods like Decision Trees, Random Forest, Support Vector Machine, and K-Nearest Neighbors have proved to be efficient tools in education-related predictions. Nevertheless, their performance may differ depending on the features of the data set, selected input variables, and parameter settings of the model used. Consequently, performing comparative research with multiple algorithms is essential to reveal the best-performing algorithm for a particular case.

The primary purpose of this work is to create and analyze several machine learning algorithms to predict students' performance. Various types of regressions, such as Linear Regression, Random Forest, K-Nearest Neighbors, Decision Trees, Gradient Boosting, Extra Trees, XGBoost, and Support Vector Regression, were examined. The efficiency of the models was measured by the coefficient of determination (R^2). The key goal of the research is to find the most efficient approach to predict student performance.

2. Related Work

Over the past few years, data mining in education has grown more popular among scientists seeking to enhance learning processes by applying advanced analysis methods. One of the most popular areas for application in educational data mining and learning analytics is predicting student academic performance. Various approaches based on machine learning technologies have been actively used to explore the educational databases and find influencing factors.

A number of works on using machine learning models to predict student academic results have been done recently. In particular, such algorithms as the Decision Tree algorithm and Random Forest were popular because of their capabilities for analyzing dependencies between input values and output ones. These approaches work well with educational data.

Moreover, Support Vector Machines and K-Nearest Neighbors were utilized in several research projects that attempted to predict student achievement. Both of these methods are capable of addressing non-linear connections between variables and have proven to be highly effective when solving classification and regression problems. Finally, another approach known as ensemble learning has been proven to enhance predictive performance, with Gradient Boosting and Extra Trees being the most common methods used to combine decision trees into one model.

Finally, more advanced machine learning techniques, namely Extreme Gradient Boosting, have gained widespread popularity due to their exceptional predictive capabilities and computational speed. XGBoost has been successfully utilized in a variety of prediction problems, including educational statistics.

Even though there have been many studies conducted in this field, it is possible that the accuracy of machine learning models could vary according to the dataset used, the approach employed for data analysis, and the way the results have been evaluated. It is crucial to carry out comparative studies, wherein different machine learning algorithms are applied to the same dataset in order to find out which models give more accurate results.

3. Dataset Description

The dataset utilized in this paper has been sourced from the Kaggle data platform where one can download datasets to support machine learning studies.

The dataset considered in this study is associated with the prediction of academic performance by considering some educational and demographic factors. There are a total of 708 student records and 10 variables in this dataset, which reflect various characteristics of students and might influence their academic performance.

Every single record refers to a certain student and reflects both numerical and categorical characteristics, which relate to study habits, learning conditions, and past academic performance of students. These characteristics include, among others, study hours per week, attendance rate, examination scores in the past, education level of parents, whether there is internet access at home, and involvement in additional activities.

The main outcome of interest is the final academic performance of students and it is represented by “Final Exam Score”. Machine learning models will be created based on the data contained in other variables.

There are a number of important variables in this dataset that offer useful information about characteristics influencing student performance. Examples include Study “Hours Per Week” reflecting the number of study hours, “Attendance Rate” showing “attendance rate of students”, and “Past Exam Scores” indicating past academic achievement.

In addition to the academic measures, the data set also contains contextual variables like Parental Education Level, Internet Access at Home, and Extracurricular Activities, which could have an indirect effect on students’ learning outcomes. Machine learning algorithms can then use these variables to explore complicated interactions between academics and external factors.

Prior to running the machine learning algorithms, the data set was carefully analyzed and preprocessed using several techniques.

Feature	Description
Study_Hours_per_Week	Number of hours the student studies per week
Attendance_Rate	Percentage of class attendance
Past_Exam_Scores	Scores obtained in previous exams
Parental_Education_Level	Education level of the student's parents
Internet_Access_at_Home	Whether the student has internet access at home
Extracurricular_Activities	Participation in activities outside the classroom

Final_Exam_Score	Final academic score obtained by the student
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4. Methodology

This section covers the dataset that is used in this research, data preprocessing techniques, ML models used for prediction, and performance measures employed to evaluate the model performance.

4.1 Dataset

This dataset is aimed at predicting the performance level of students depending on a variety of parameters in the learning environment. This dataset includes multiple records pertaining to individual students, as well as a number of features pertaining to the academic learning environment.

This dataset comprises several different variables that could affect student performance such as study time, attendance, past academic record, among others. The target variable is the final performance of each student predicted by the algorithms.

The dataset was further subdivided into a training subset and a testing subset. The former was used for training the algorithms, while the latter was employed for assessing the accuracy of these predictions.

4.2 Data Preprocessing

Various processes of preprocessing the data before feeding it to the machine learning models were done. Preprocessing of data in machine learning is an essential process because it helps to ensure that the data is ready for the machine learning algorithm.

Missing data was checked in the data set, and where applicable, suitable techniques were employed to deal with it. The second step involves feature scaling to ensure that the values of all features are standardized. This step is crucial in cases of algorithms such as K-Nearest Neighbors and Support Vector Regression which depend on the scale of the features.

The next step was splitting of the data set into training and test data set.

4.3 Machine Learning Models

To evaluate the effectiveness of machine learning techniques for predicting student performance, multiple regression algorithms were implemented and compared in this study.

The models used in this research include:

- Linear Regression
- Random Forest Regressor
- K-Nearest Neighbors Regressor
- Radius Neighbors Regressor
- Decision Tree Regressor
- Extra Tree Regressor
- Extra Trees Regressor
- Gradient Boosting Regressor
- Extreme Gradient Boosting (XGBoost)
- Support Vector Regression (SVR) with different kernels
- Nu-Support Vector Regression (NuSVR)

The reason why these models have been chosen is that they are diverse examples of machine learning algorithms such as linear models, decision tree models, ensemble learning techniques, and kernel-based models. Through comparisons of these algorithms, the accuracy of the prediction can be found out easily.

4.4 Model Evaluation

For assessing the prediction performance of the developed models, the coefficient of determination, R^2 , was adopted as the key performance index. This performance index gives information regarding the closeness between the predicted and the actual values.

The coefficient of determination can be calculated using the formula below:

$$R^2 = 1 - \frac{\sum (y_i - \bar{y})^2}{\sum (y_i - \hat{y}_i)^2}$$

where y_i is the actual data values, $\hat{y}_i$ is the predicted data values, while $\bar{y}$ is the mean of the actual data values. The higher the R^2 value, the better the prediction performance of the developed model.

5. Experiments and Results

5.1 Experimental Setup

In order to analyze the performance of the proposed machine learning models, various regression techniques were applied using Python programming language and Scikit-Learn Machine Learning Package. Both models were trained on the training dataset and tested on the test dataset in order to estimate their generalization capabilities.

Each model was trained based on the same dataset and the same preprocessing methods in order to allow unbiased comparisons between different approaches. The performance of each model was estimated by calculating the coefficient of determination (R^2).

5.2 Model Performance Comparison

After training the models, the predictive performance of each algorithm was calculated using the R^2 metric. Table 1 presents the performance of the implemented models.

Model	R^2 Score
Linear Regression	0.7769
Random Forest	0.7125
K-Nearest Neighbors	0.6845
Radius Neighbors	0.5153
Decision Tree	0.4213
Extra Tree	0.4706
Extra Trees	0.7184
Gradient Boosting	0.6901
XGBoost	0.6278
SVR (RBF Kernel)	0.7619
SVR (Linear Kernel)	0.7759
SVR (Polynomial Kernel)	0.7434
NuSVR	0.7396

Based on the findings, various machine learning models have diverse performances in terms of prediction. The ensemble machine learning techniques, which include random forest, gradient boosting, and XGBoost, tend to perform well when it comes to predicting because of the incorporation of many decision trees.

5.3 Visualization of Model Performance

In order to present a more effective comparison between the two machine learning algorithms, the results for their respective R^2 values were plotted using a graph.

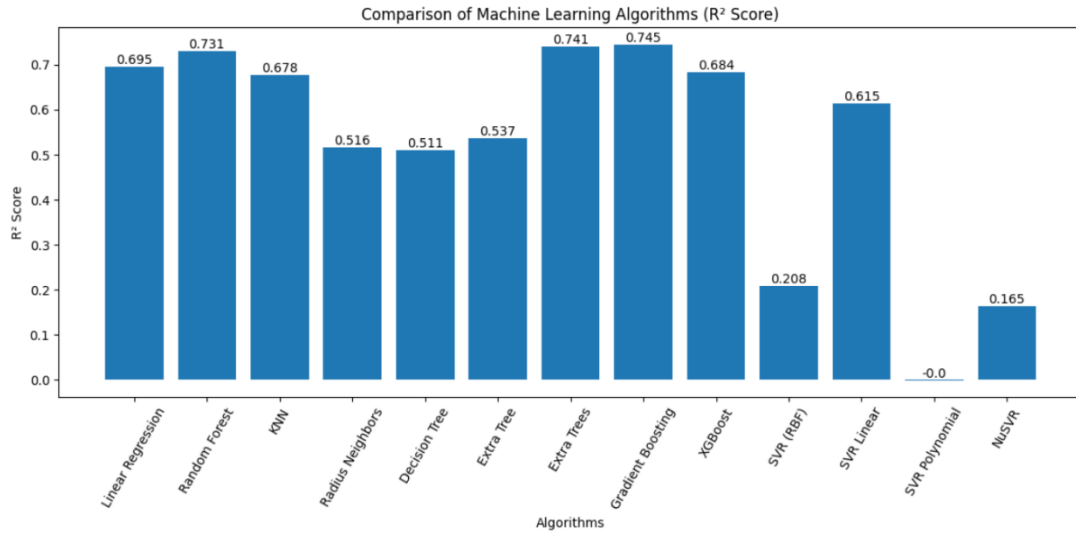


Figure 1. Comparison of machine learning algorithms based on R^2 performance.

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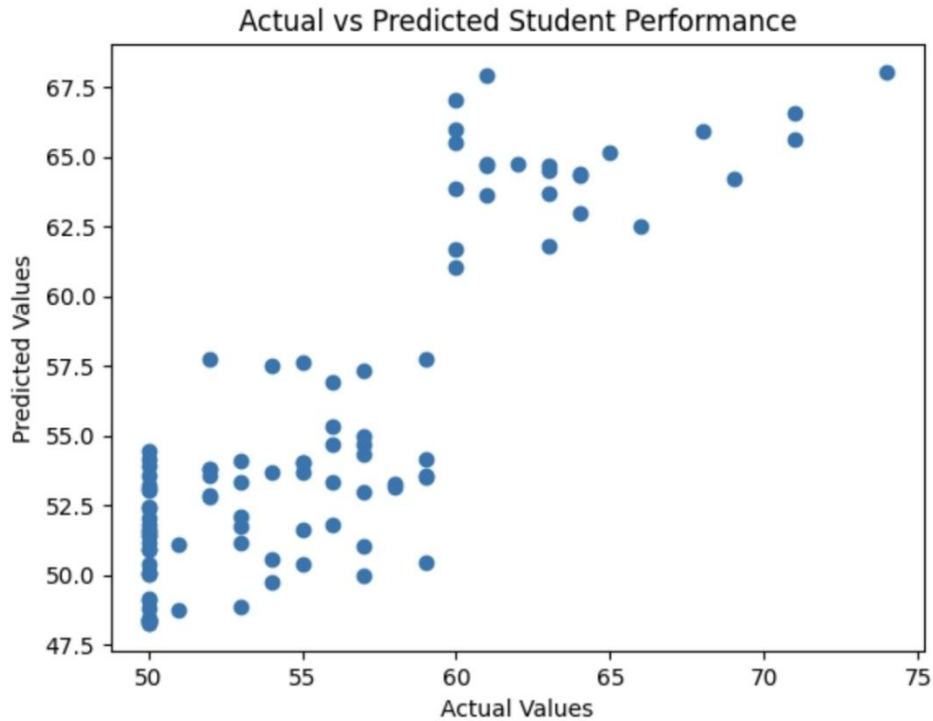


Figure 2. Actual vs Predicted Student Performance.

This is a scatter plot showing the actual test marks of the students against the predicted values from the machine learning algorithm used to generate the predictions. The degree of closeness of the points to the diagonal is an indication of the accuracy of prediction.

5.4 Discussion of Results

From the experimental analysis conducted, it is noted that there exists a disparity among various machine learning algorithms when it comes to accurately predicting student performance in class. Algorithms such as Random Forest, Gradient Boosting, and XGBoost have been seen to have greater accuracy due to their ensemble nature, which allows for lower error rates while making predictions.

Linear Regression, on the other hand, can sometimes fail to detect any relationship between variables since it is simple. Also, algorithms like Support Vector Regression and K-Nearest Neighbors can prove effective, depending on the type of dataset used and the choice of parameters used.

In summary, machine learning algorithms can predict student performance accurately.

7. Conclusion

This study focuses on analyzing the implementation of machine learning for making predictions on student performance based on an academic and non-academic dataset. A variety of multiple regression models were designed and compared, namely Linear Regression, Random Forest, KNN, Decision Tree, Extra Trees, Gradient Boosting, XGBoost, and Support Vector Regression.

Based on the analysis of the results obtained, one can conclude that machine learning approaches have a high level of effectiveness in finding connections between factors influencing educational achievements and student performance. It should be noted that the comparative analysis of different algorithms showed differences in terms of prediction accuracy, which proves the necessity of choosing the right algorithm for each particular educational task. As far as ensemble techniques (Random Forest, Gradient Boosting, XGBoost) showed good results in the research, it can be stated that this type of algorithms should be applied to improve student performance prediction.

The outcomes of this study prove the significance of machine learning techniques in helping educational institutions and teachers to identify problematic students who need additional help.

Further improvements in predictive performance can be achieved in future work by employing more datasets, complex feature extraction methods, and deep learning architectures. Also, using live data collected via learning management systems will be useful for better understanding learners' behavior and performance.

For reproducibility of the experimental results presented in this paper, the source code of the implemented machine learning models and details on the experimental setup are provided in the following GitHub repository:

Code Repository:

<https://github.com/ReineNassour/Machine-Learning-Models-for-Predicting-Student-Academic-Performance>

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